

## C1 · Parsing-Retrieval Disconnect

### Candidate parsers



Qwen3-VL



MinerU



Marker



PaddleOCR

### Cleaner-looking $\neq$ better retrieval

| parser      | Boundary Clarity | Hit@1       |
|-------------|------------------|-------------|
| MinerU      | 0.72 (highest)   | <b>0.20</b> |
| Prod (ours) | 0.61             | <b>0.55</b> |

parser choice alone swings Hit@1 by **2.8×**  
appearance metrics measure formatting, not content  
(anti-correlation  $r = -0.81$ ,  $n = 5$ )

## C2 / C3 · Diagnose + Select

### C2 · Coverage diagnostic (no retriever)

where does each gold answer land?



covered

absent

absent = answer never made it into the parser output  
**(parser fault)**  
constant across all 8 chunkers  $\rightarrow$  chunking can't recover it

### C3 · RCPS selection protocol (no training)

held-out Q-A probe, scored over 3 retrievers:



BGE-M3



mE5



Qwen3-Emb

**$\rightarrow$  rank parsers & chunkers by retrieval**

## C4 · Bounded Parser Training

### C4 · Parser-side training

best-of-K parses from Prod, ranked two ways:

**RADP-Distill — edit-distance  $\rightarrow$  GT**

**RADP-DPO — page-local RCPS**

### Result

Hit@5 gain over the untuned parser:

**RADP-Distill**



**+1.22 pp**

**RADP-DPO**



**+0.85 pp**

$\approx$  **overlapping CIs  $\rightarrow$  equal gain**  
**retrieval reward unnecessary; lever = fidelity distillation**